No takers for ver 3?
Perhaps the backward incompatibility of the latest version of Python is the reason why it's not being adopted as easily

What's in a name?
The name python is a reference to the infamous comedy group Monty Python's Flying Circus. Looks like the developers of the language had a funny bone

Python
- a quick and dirty tutorial

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Whether you have used it or not, Python is probably a language you all have heard of. The reason is simple, this language has friends in high places.

Python is a favoured language at Google, and was one of the first languages supported in its Google App Engine platform for developing web applications. It is also the language in which Google choose to write its GoogleCL command line interface for Google services. Many organisation use Python both internally and in shipping products. The popular Civilization IV game by Firaxis for example, makes extensive use of Python for the game mechanics; Industrial Light and Magic too use Python, as does Yahoo. Blender, possibly the most popular free and open source software for creating and editing 3D content, makes significant use of Python. While Firefox itself isn't written in Python, its build system makes extensive use of Python.

So why so much love for Python? Well, because it is fast, it is free and open source, and best of all it is easy to learn. Python is portable, and is available for multiple platforms, Windows, Linux, Mac OS, and is embeddable in other products. For someone new to programming, Python is a great language to start with, and will likely stay with you even as you learn other languages.

Python is a high level language, which means most operations you wish to perform have libraries and can be handled in just a few lines. You can, for example, have a simple web server up and running in under 100 lines of code, thanks to the HTTPServer class.

Python is extensible, and one will find thousands of modules which extend its functionality. Python is scalable, considering that YouTube, is almost entirely Python! Most of the modules that come with Python are actually written in C / C++ so they run at native speeds, and your programs generally run faster than those of other interpreted programming languages.

Python is good at interfacing with other programming languages such as Java and C++ making it a great language for programming higher level features of an application, while keeping performance-critical functions in native code. This enables Python developers to easily use C / C++ libraries such as Qt / GTK+ and other

Python is also peculiar. Unlike most programming languages, where white-spaces and tabs are used simply for code formatting and discarded otherwise, Python uses these elements as part of the code structure. Python uses spacing instead of '{' brackets or text to denote blocks of code.

In one sentence, Python is a fast, interpreted, Object-Oriented, Dynamic, free, Open Source, powerful, extensible, modular, high-level, portable programming language that is easy to learn, and which has a powerful library, good documentation, a large number of third party modules, and a sense of humour.

Learning Python
Whether you are simply looking for a language to automate simple tasks or, process data in bulk, or if you want to prototype or write an entire application Python is a great tool for the job.

Since Python enforces code indentation, and makes it part of the language, Python code is quite easy to understand. Look at the following code sample:

```
a, b = 0, 1
while a < 1000000:
    a, b = b, a+b
    print(a)
```

By looking at this program it is quite easy to understand what it does. The first line simultaneously assigns the value 0 to a, and 1 to b. The second line tells us that the subsequent code needs to be repeated while the condition following it holds true. The third line is clear in that it prints the value of the variable b. The final line, sets the value of a to b and the value of b to a+b.

Some of you may be aware of the mathematical significance of this code, but for those who aren't, this code will print the Fibonacci sequence. The Fibonacci sequence is a series of numbers where each number is the sum of the previous two numbers in the series. The first two numbers are conventionally both 1.

However, let us make one small change in the program:

```
a, b = 0, 1
while a < 1000000:
    a, b = b, a+b
print(a)
```

Now as you can see, the indentation of print(a) has been decreased, putting it in line with the while loop instead of its contents. This small change stands to completely change what out code does. In this new code, instead of printing all Fibonacci numbers up till a certain point, it will print only the first number of the Fibonacci sequence which is greater than the specified limit. This is because, instead of being printed at each execution of the loop, now this number will only execute once, at the end.

You may have noticed that we did not need to declare any of the variables before assigning values to them.

Numbers in Python
Python is very powerful at handling numbers. In the previous code sample for example, you would have to add considerably more zeros to 1000000 before you will actually notice the program taking time in performing the operation.

You can do simple mathematical operation with ease. Operations such as +, -, * and / work as expected. Like most other languages Python also includes the remainder or mod function %. However, Python in addition supports a '//' operation, which is an integer division operation. It will return the integer result for the operation while discarding the fraction. For example:

```
>>> 19/4
```